# BLAKKBOX UNIVERSAL DENSO DIESEL MASTER WORKFLOW

## UNIVERSAL OEM-PRESERVING CALIBRATION FRAMEWORK

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# CORE PHILOSOPHY

FINAL = ORIGINAL + FILTERED_DELTA

Primary operating principle:

- ORIGINAL remains read-only
- Only modified calibration regions are processed
- Untouched OEM regions remain byte-identical
- Preserve executable integrity
- Preserve OEM interpolation behavior
- Preserve torque hierarchy structure
- Preserve combustion stability
- Preserve OEM fail-safe logic

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# SUPPORTED MANUFACTURERS

## TOYOTA DENSO

- 1KD-FTV
- 2KD-FTV
- 1GD-FTV
- 2GD-FTV
- 1VD-FTV

## MITSUBISHI DENSO

- 4D56
- 4M41
- 4N15

## NISSAN DENSO

- YD25

- ZD30
- E63

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# SUPPORTED CPU ARCHITECTURES

- SH7055
- SH7058
- SH7059
- SH72531
- SH72543
- RH850
- MPC5xx
- TriCore

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# SUPPORTED BIN SIZES

- 320KB
- 368KB
- 512KB
- 716KB
- 896KB
- 1MB
- 1.5MB
- 2MB
- 2.5MB
- 3MB
- 4MB

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# CORE EXECUTION PRINCIPLE

1. Prove layout first
2. Prove structure second
3. Prove calibration meaning last

Never tune before structural verification.

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# BASE EXECUTION RULES

## ORIGINAL.bin

READ ONLY

## MOD.bin

ACTIVE WORKSPACE

NEVER:

- overwrite ORIGINAL
- normalize entire ROM
- rewrite untouched regions
- checksum-correct during processing
- alter executable regions blindly

ALWAYS PRESERVE:

- DTC masks
- EGR regions
- watchdog logic
- bootloader structure
- OEM fail-safe systems
- unused OEM regions

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# FIRMWARE NORMALIZATION

SUPPORTED INPUTS:

- BIN
- HEX
- S19
- MOT
- ELF

NORMALIZATION RULES:

- convert to flat binary
- preserve address continuity
- preserve byte alignment
- reject shifted binaries
- reject displaced geometry

# MANDATORY PRE-CHECKS

VERIFY:

- SW ID
- exact ROM size
- geometry continuity
- address alignment
- no offset drift
- architecture plausibility
- executable preservation

REJECT:

- corrupted structure
- shifted binaries
- executable contamination
- ROM size mismatch
- invalid geometry

# ARCHITECTURE FINGERPRINTING

Detect using:

- reset vectors
- opcode signatures
- pointer spacing
- interrupt layouts
- memory segmentation

Segment:

- bootloader
- application
- calibration
- NVM
- protected regions

# ENDIANNESS VALIDATION

VERIFY:

- pointer plausibility
- structure alignment
- stable decoding
- axis continuity
- checksum layout consistency

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# ENTROPY CLASSIFICATION

LOW ENTROPY:

- calibration surfaces
- axis structures

MEDIUM ENTROPY:

- executable regions

HIGH ENTROPY:

- compressed/protected regions

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# ROM INTEGRITY ENGINE

VERIFY:

- SHA256 integrity
- executable preservation
- geometry preservation
- region continuity
- offset stability
- contiguous calibration structures

EXPECTED:

- structured grouped modifications
- stable executable layout
- smooth calibration surfaces

REJECT:

- fragmented overwrite behavior
- blanket modification regions
- executable contamination
- geometry corruption

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# SMART DELTA EXTRACTION

DELTA = MOD - ORIGINAL

ONLY modified regions are processed.

Untouched OEM regions remain byte-identical.

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# MODIFIED REGION DETECTION

Detect:

- contiguous modified clusters
- synchronized map structures
- grouped calibration surfaces
- limiter synchronization

Classify:

- structured calibration edits
- fragmented corruption
- executable contamination

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# MAP DETECTION HEURISTICS

Detect:

- rectangular surfaces
- monotonic axes
- smooth interpolation
- grouped surfaces
- neighbor continuity

Validate:

- row smoothness
- column smoothness
- axis continuity
- engineering plausibility

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# AXIS VALIDATION

VERIFY:

- monotonic progression
- stable scaling
- neighboring continuity
- interpolation stability

REJECT:

- broken progression
- jagged transitions
- unstable gradients

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# TORQUE HIERARCHY VALIDATION

VERIFY:

- driver demand consistency
- limiter synchronization
- torque monitor alignment
- airflow/fuel coordination
- vane synchronization

Reject:

- abrupt torque spikes
- isolated fuel expansion
- uncontrolled boost behavior

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# MULTI-BANK SYNCHRONIZATION

VERIFY:

- mirrored map consistency
- synchronized limiter scaling
- coordinated airflow/fuel logic
- bank continuity

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# DELTA DENSITY CLUSTER ANALYSIS

Analyze:

- modification density
- grouped calibration regions
- synchronized torque structures
- executable isolation

Typical ranges:

STAGE 1: 5,000–16,000 changed bytes

STAGE 2: 15,000–35,000 changed bytes

SPORT / SP V2: 25,000–55,000 changed bytes

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# CALIBRATION PROFILE DETECTION

Classify:

- OEM+
- Stage 1
- Stage 2
- Sport
- SP V2
- ECO

Using:

- delta amplitude
- density distribution
- limiter expansion

• airflow scaling behavior

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# BLAKKBOX DELTA LOGIC

Δ0–5: KEEP

Δ5–8: ×0.80

Δ>8: ×0.55

Additional filtering:

- spike suppression
- jagged transition smoothing
- neighboring continuity preservation
- torque spike prevention
- rail overshoot prevention
- vane transition stabilization

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# AGGRESSIVE DELTA ENGINE

Δ0–5: ×1.15–1.25

Δ5–10: ×1.22–1.35

Δ>10: ×1.30–1.42

Additional restraint:

- suppress spikes above Δ40
- stabilize neighboring gradients
- preserve combustion safety

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# ECONOMY DELTA ENGINE

Δ0–5: ×0.88–0.92

Δ5–10: ×0.74–0.82

Δ>10: ×0.62–0.72

Prioritize:

- low smoke
- smooth transient
- rail stability
- low EGT behavior

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# INTERPOLATION PRESERVATION

DO:

- smooth jagged transitions
- preserve neighboring trends
- preserve gradient continuity
- preserve map geometry

DO NOT:

- create cliffs
- create spikes
- break axis continuity
- generate unstable transitions

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# COMBUSTION REFINEMENT STRATEGY

TARGET AREAS:

- startup
- idle
- low load
- 1000–2500 RPM
- transient response
- WOT transition

OBJECTIVES:

- reduce smoke
- reduce diesel knock
- reduce clatter
- stabilize combustion
- stabilize transient fueling

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# AIRFLOW SYNCHRONIZATION

PRIORITY: AIRFLOW > FUEL

NEVER:

- excessive IQ before airflow
- unstable vane transitions
- abrupt torque delivery
- uncontrolled boost rise

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# RAIL PRESSURE SAFETY

VERIFY:

- smooth rail transitions
- stable pressure gradients
- synchronized IQ relationship

REJECT:

- abrupt rail spikes
- unstable transient pressure
- isolated pressure escalation

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# HEX INTELLIGENCE ENGINE

Generate:

- modified region analysis
- entropy scoring
- binary heatmaps
- probable axis detection
- continuity plots
- grouped region visualization

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# CONFIDENCE SCORING

FACTORS:

- pointer validity
- axis monotonicity
- smoothness
- alignment correctness
- checksum compatibility
- grouped continuity

CONFIDENCE LEVELS:

90–100: CONFIRMED

75–89: HIGHLY PROBABLE

50–74: PROBABLE

<50: WEAK

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# ADVANCED ANALYSIS METHODS

- byte-level comparison
- delta density analysis
- torque-structure interpretation
- region clustering
- interpolation plotting
- neighboring gradient analysis
- executable isolation analysis

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# REGRESSION VALIDATION

VERIFY:

- deterministic output
- stable parser behavior
- stable grouping behavior
- stable interpolation continuity
- no geometry corruption
- no executable contamination
- no offset drift

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# FINAL EXPORT RULES

EXPORT:

- FINAL BIN
- CSV REPORT
- PDF ANALYSIS

CHECKSUM: DO NOT CORRECT DURING PROCESSING

FINAL CHECKSUM: MANUAL BITEDIT CHECKSUM ONLY

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# FINAL CALIBRATION PHILOSOPHY

Best-performing calibrations are not the most aggressive.

Best-performing calibrations maintain:

- combustion stability
- torque consistency
- airflow synchronization
- OEM structural integrity
- smooth interpolation
- thermal control
- mechanical sympathy

GOOD CALIBRATION:

- combustion efficient
- smoke controlled
- torque stable
- OEM structured
- interpolation clean

BAD CALIBRATION:

- smoke rich
- torque spiky
- unstable transient
- broken interpolation
- executable contamination